



# **GCSE MARKING SCHEME**

**SUMMER 2022**

**GCSE  
CHEMISTRY – UNIT 1  
3410U10-1 AND 3410UA0-1**

## **INTRODUCTION**

This marking scheme was used by WJEC for the 2022 examination. It was finalised after detailed discussion at examiners' conferences by all the examiners involved in the assessment. The conference was held shortly after the paper was taken so that reference could be made to the full range of candidates' responses, with photocopied scripts forming the basis of discussion. The aim of the conference was to ensure that the marking scheme was interpreted and applied in the same way by all examiners.

It is hoped that this information will be of assistance to centres but it is recognised at the same time that, without the benefit of participation in the examiners' conference, teachers may have different views on certain matters of detail or interpretation.

WJEC regrets that it cannot enter into any discussion or correspondence about this marking scheme.

**WJEC GCSE CHEMISTRY**

**UNIT 1 – CHEMICAL SUBSTANCES, REACTIONS AND ESSENTIAL RESOURCES**

**SUMMER 2022 MARK SCHEME**

**GENERAL INSTRUCTIONS**

Marking rules

All work should be seen to have been marked.

Marking schemes will indicate when explicit working is deemed to be a necessary part of a correct answer.

Crossed out responses not replaced should be marked.

Credit will be given for correct and relevant alternative responses which are not recorded in the mark scheme.

Extended response question

A level of response mark scheme is used. Before applying the mark scheme please read through the whole answer from start to finish. Firstly, decide which level descriptor matches best with the candidate's response: remember that you should be considering the overall quality of the response. Then decide which mark to award within the level. Award the higher mark in the level if there is a good match with both the content statements and the communication statements.

Marking abbreviations

The following may be used in marking schemes or in the marking of scripts to indicate reasons for the marks awarded.

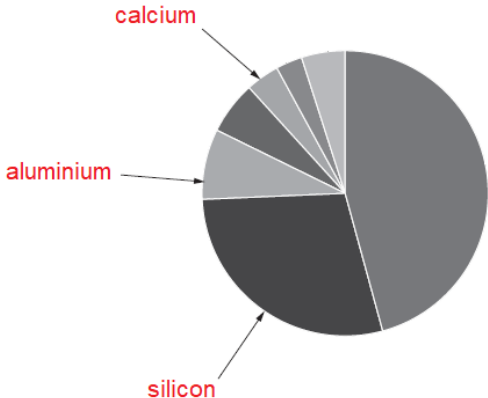
|     |   |                       |
|-----|---|-----------------------|
| cao | = | correct answer only   |
| ecf | = | error carried forward |
| bod | = | benefit of doubt      |

## Foundation Tier only questions

| Question |     |      |  | Marking details                                                                                                                                                                                                                                        |                |         |          | Marks available |         |         |          |                     |      |  |   |          |                |   |  |  |   |  |   |  |  |
|----------|-----|------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------|----------|-----------------|---------|---------|----------|---------------------|------|--|---|----------|----------------|---|--|--|---|--|---|--|--|
|          |     |      |  |                                                                                                                                                                                                                                                        |                |         |          | AO1             | AO2     | AO3     | Total    | Maths               | Prac |  |   |          |                |   |  |  |   |  |   |  |  |
| 1        | (a) | (i)  |  | <table><tr><td>Substance</td><td>Formula</td><td>Element</td><td>Compound</td></tr><tr><td>potassium hydroxide</td><td>KOH</td><td></td><td>✓</td></tr><tr><td>hydrogen</td><td>H<sub>2</sub></td><td>✓</td><td></td></tr></table>                     |                |         |          | Substance       | Formula | Element | Compound | potassium hydroxide | KOH  |  | ✓ | hydrogen | H <sub>2</sub> | ✓ |  |  | 2 |  | 2 |  |  |
|          |     |      |  | Substance                                                                                                                                                                                                                                              | Formula        | Element | Compound |                 |         |         |          |                     |      |  |   |          |                |   |  |  |   |  |   |  |  |
|          |     |      |  | potassium hydroxide                                                                                                                                                                                                                                    | KOH            |         | ✓        |                 |         |         |          |                     |      |  |   |          |                |   |  |  |   |  |   |  |  |
|          |     |      |  | hydrogen                                                                                                                                                                                                                                               | H <sub>2</sub> | ✓       |          |                 |         |         |          |                     |      |  |   |          |                |   |  |  |   |  |   |  |  |
|          |     |      |  |                                                                                                                                                                                                                                                        |                |         |          |                 |         |         |          |                     |      |  |   |          |                |   |  |  |   |  |   |  |  |
|          |     |      |  |                                                                                                                                                                                                                                                        |                |         |          |                 |         |         |          |                     |      |  |   |          |                |   |  |  |   |  |   |  |  |
|          |     |      |  |                                                                                                                                                                                                                                                        |                |         |          |                 |         |         |          |                     |      |  |   |          |                |   |  |  |   |  |   |  |  |
|          |     | (ii) |  | fizzing (1)<br>potassium floats (1)                                                                                                                                                                                                                    |                |         |          | 2               |         |         | 2        |                     | 2    |  |   |          |                |   |  |  |   |  |   |  |  |
|          | (b) |      |  | 4<br>accept any method of identifying correct answer                                                                                                                                                                                                   |                |         |          |                 | 1       |         | 1        | 1                   |      |  |   |          |                |   |  |  |   |  |   |  |  |
|          | (c) |      |  | <div><div><div>white</div></div><div><div>lithium</div><div>red</div></div><div><div>sodium</div><div>green</div></div><div><div>barium</div><div>blue</div></div><div><div></div><div>yellow</div></div></div> <p>award (1) for each correct line</p> |                |         |          | 2               |         |         | 2        |                     | 2    |  |   |          |                |   |  |  |   |  |   |  |  |
|          |     |      |  |                                                                                                                                                                                                                                                        |                |         |          |                 |         |         |          |                     |      |  |   |          |                |   |  |  |   |  |   |  |  |
|          |     |      |  | Question 1 total                                                                                                                                                                                                                                       |                |         |          | 4               | 3       | 0       | 7        | 1                   | 4    |  |   |          |                |   |  |  |   |  |   |  |  |

| Question |     |  |  | Marking details                                    | Marks available |          |          |          |          |          |
|----------|-----|--|--|----------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |  |  |                                                    | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 2        | (a) |  |  | sulfur dioxide, SO <sub>2</sub> <b>B</b> (1)       |                 |          |          |          |          |          |
|          |     |  |  | ethene, C <sub>2</sub> H <sub>4</sub> <b>A</b> (1) |                 | 2        |          | 2        |          |          |
|          | (b) |  |  | carbon dioxide                                     |                 | 1        |          | 1        |          |          |
|          |     |  |  | neutral answer – CO <sub>2</sub>                   |                 |          |          |          |          |          |
|          |     |  |  | <b>Question 2 total</b>                            | <b>0</b>        | <b>3</b> | <b>0</b> | <b>3</b> | <b>0</b> | <b>0</b> |

| Question |     |      |    | Marking details                                                                                                                                                                                                                        | Marks available |          |          |          |          |          |
|----------|-----|------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |      |    |                                                                                                                                                                                                                                        | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 3        | (a) |      |    | 7                                                                                                                                                                                                                                      |                 | 1        |          | 1        |          |          |
|          | (b) |      |    | Cl <sub>2</sub>                                                                                                                                                                                                                        |                 | 1        |          | 1        |          |          |
|          | (c) | (i)  |    | glowed less brightly than iodine <input type="checkbox"/><br>glowed less brightly than chlorine <input checked="" type="checkbox"/><br>glowed more brightly than chlorine <input type="checkbox"/>                                     |                 | 1        |          | 1        |          | 1        |
|          |     | (ii) | I  | FeBr <sub>3</sub>                                                                                                                                                                                                                      |                 | 1        |          | 1        |          |          |
|          |     |      | II | iron bromide      ignore any bracketed numbers                                                                                                                                                                                         |                 | 1        |          | 1        |          |          |
|          | (d) |      |    | to disinfect skin before surgery <input checked="" type="checkbox"/><br>to make coloured fireworks <input type="checkbox"/><br>to sterilise swimming pools <input type="checkbox"/><br>to fill party balloons <input type="checkbox"/> | 1               |          |          | 1        |          |          |
|          |     |      |    | <b>Question 3 total</b>                                                                                                                                                                                                                | <b>1</b>        | <b>5</b> | <b>0</b> | <b>6</b> | <b>0</b> | <b>1</b> |

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                          | Marks available |     |     |       |       |      |
|----------|-----|-------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                                                                          | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 4        | (a) | (i)   |  |  <p>award (2) for all correct<br/>award (1) for 1 or 2 correct</p>                                                                                                                                      |                 |     | 2   | 2     |       |      |
|          |     | (ii)  |  | $\frac{1}{4}$                                                                                                                                                                                                                                                                            |                 |     | 1   | 1     | 1     |      |
|          |     | (iii) |  | <p>most metals have higher melting points than gold <input type="checkbox"/></p> <p>most metals are magnetic <input type="checkbox"/></p> <p>most metals are more reactive than gold <input checked="" type="checkbox"/></p> <p>most metals are radioactive <input type="checkbox"/></p> | 1               |     |     | 1     |       |      |

| Question |     |      |  | Marking details                                                                                                                                                                               | Marks available |          |          |          |          |          |
|----------|-----|------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |      |  |                                                                                                                                                                                               | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
|          | (b) | (i)  |  | plates move apart / separate (1)<br>magma / molten rock moves upwards (1)<br>cools / solidifies / crystallises / hardens (1)<br>references to new land / igneous rock / volcanoes are neutral | 3               |          |          | 3        |          |          |
|          |     | (ii) |  | constructive                                                                                                                                                                                  | 1               |          |          | 1        |          |          |
|          |     |      |  | <b>Question 4 total</b>                                                                                                                                                                       | <b>5</b>        | <b>0</b> | <b>3</b> | <b>8</b> | <b>1</b> | <b>0</b> |



| Question                |     |       |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                               | Marks available         |     |               |       |                       |      |                 |   |                  |  |                    |   |  |  |   |   |  |   |
|-------------------------|-----|-------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-----|---------------|-------|-----------------------|------|-----------------|---|------------------|--|--------------------|---|--|--|---|---|--|---|
|                         |     |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                               | AO1                     | AO2 | AO3           | Total | Maths                 | Prac |                 |   |                  |  |                    |   |  |  |   |   |  |   |
| 5                       | (a) | (i)   |  | <b>D <u>B</u> E A C</b>                                                                                                                                                                                                                                                                                                                                                                                                       |                         |     | 1             | 1     |                       | 1    |                 |   |                  |  |                    |   |  |  |   |   |  |   |
|                         |     | (ii)  |  | magnesium                                                                                                                                                                                                                                                                                                                                                                                                                     | 1                       |     |               | 1     |                       |      |                 |   |                  |  |                    |   |  |  |   |   |  |   |
|                         |     | (iii) |  | <table><tr><td>volume of soap solution</td><td>✓</td></tr><tr><td>type of water</td><td></td></tr><tr><td>type of soap solution</td><td>✓</td></tr><tr><td>volume of water</td><td>✓</td></tr><tr><td>height of lather</td><td></td></tr><tr><td>width of test tube</td><td>✓</td></tr></table><br>award (2) for all 4 correct with no other ticks<br><br>award (1) for any 2 or 3<br><br>deduct (1) per incorrect box ticked | volume of soap solution | ✓   | type of water |       | type of soap solution | ✓    | volume of water | ✓ | height of lather |  | width of test tube | ✓ |  |  | 2 | 2 |  | 2 |
| volume of soap solution | ✓   |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                               |                         |     |               |       |                       |      |                 |   |                  |  |                    |   |  |  |   |   |  |   |
| type of water           |     |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                               |                         |     |               |       |                       |      |                 |   |                  |  |                    |   |  |  |   |   |  |   |
| type of soap solution   | ✓   |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                               |                         |     |               |       |                       |      |                 |   |                  |  |                    |   |  |  |   |   |  |   |
| volume of water         | ✓   |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                               |                         |     |               |       |                       |      |                 |   |                  |  |                    |   |  |  |   |   |  |   |
| height of lather        |     |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                               |                         |     |               |       |                       |      |                 |   |                  |  |                    |   |  |  |   |   |  |   |
| width of test tube      | ✓   |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                               |                         |     |               |       |                       |      |                 |   |                  |  |                    |   |  |  |   |   |  |   |
|                         | (b) | (i)   |  | 54°C                  no tolerance                                                                                                                                                                                                                                                                                                                                                                                            |                         | 1   |               | 1     | 1                     |      |                 |   |                  |  |                    |   |  |  |   |   |  |   |
|                         |     | (ii)  |  | solubility of both increases as the temperature increases (1)<br><br>award (1) for any of following<br>(solubility of) copper(II) sulfate increases much more (than that of sodium chloride)<br>(solubility of) sodium chloride increases much less (than that of copper(II) sulfate)<br>(solubility of) copper(II) sulfate increases a lot and that of sodium chloride increases only slightly<br><br>comparison needed      |                         | 2   |               | 2     |                       |      |                 |   |                  |  |                    |   |  |  |   |   |  |   |

| Question |     |       |  | Marking details                                                                                                                        | Marks available |          |          |           |          |          |
|----------|-----|-------|--|----------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |     |       |  |                                                                                                                                        | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
|          |     | (iii) |  | 290 g (2)<br><br>if incorrect award (1) for either of following<br><br>29 g (correct reading from graph)<br>any value multiplied by 10 |                 | 2        |          | 2         | 2        |          |
|          | (c) | (i)   |  | 1                                                                                                                                      |                 | 1        |          | 1         |          |          |
|          |     | (ii)  |  | 174 (2)<br><br>if incorrect award (1) for either of following<br><br>$(39 \times 2) + 32 + (4 \times 16)$<br><br>2 K + 1 S + 4 O       |                 | 2        |          | 2         | 2        |          |
|          |     |       |  | <b>Question 5 total</b>                                                                                                                | <b>1</b>        | <b>8</b> | <b>3</b> | <b>12</b> | <b>5</b> | <b>3</b> |

| Question |     |       |  | Marking details                                                                                                                                                                                                       | Marks available |          |          |          |          |          |
|----------|-----|-------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |       |  |                                                                                                                                                                                                                       | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 6        | (a) | (i)   |  | atomic number $\Rightarrow$ 5 (1)<br>mass number $\Rightarrow$ 11 (1)                                                                                                                                                 |                 | 2        |          | 2        |          |          |
|          |     | (ii)  |  | <p>accept 2,3 as a written alternative</p>                                                                                                                                                                            |                 | 1        |          | 1        |          |          |
|          |     | (iii) |  | equal numbers of protons and electrons (1)<br>protons are positive and electrons are negative / protons and electrons have opposite charges (1)<br>neutral answers<br>any reference to neutrons<br>charges cancel out | 2               |          |          | 2        |          |          |
|          | (b) | (i)   |  | nitrogen accept N / N <sub>2</sub>                                                                                                                                                                                    |                 | 1        |          | 1        |          |          |
|          |     | (ii)  |  | 5 electrons in <u>outer</u> shell / orbit                                                                                                                                                                             |                 | 1        |          | 1        |          |          |
|          |     | (iii) |  | 2 (electron) shells / orbits                                                                                                                                                                                          |                 | 1        |          | 1        |          |          |
|          |     |       |  | <b>Question 6 total</b>                                                                                                                                                                                               | <b>2</b>        | <b>6</b> | <b>0</b> | <b>8</b> | <b>0</b> | <b>0</b> |

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks available |          |          |           |          |          |
|----------|-----|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
| 7        | (a) |  |  | award (2) for 6 correct points (tolerance $\pm\frac{1}{2}$ square)<br>award (1) for any 4 or 5 correct points<br><br>award (1) for straight line through points<br>does not need to be drawn to origin                                                                                                                                                                                                                                                                                                                                                       |                 | 2        | 1        | 3         | 3        |          |
|          | (b) |  |  | award (2) for high-level quantitative description <ul style="list-style-type: none"> <li>as the concentration doubles, the volume of gas doubles</li> <li>concentration and volume of gas are directly proportional</li> </ul> award (1) for lower-level description <ul style="list-style-type: none"> <li>as the concentration increases, the volume of gas increases</li> <li>concentration and volume are proportional</li> <li>concentration and volume are directly correlated</li> <li>concentration and volume have a linear relationship</li> </ul> |                 |          | 2        | 2         |          | 2        |
|          | (c) |  |  | more (1)<br><br>collide (1)<br><br>gas (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2               | 1        |          | 3         |          | 1        |
|          | (d) |  |  | award (1) each for any <b>two</b> of following <ul style="list-style-type: none"> <li>increase temperature / warm / heat / hotter</li> <li>increase surface area (of chalk) / smaller pieces / cut chalk up / powder chalk [do not accept smaller surface area]</li> <li>(add) catalyst (1)</li> </ul> award (1) for 'change' surface area <u>and</u> temperature with no reference to 'increase' if no other mark awarded                                                                                                                                   |                 |          | 2        | 2         |          | 2        |
|          |     |  |  | <b>Question 7 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>2</b>        | <b>3</b> | <b>5</b> | <b>10</b> | <b>3</b> | <b>5</b> |

| Question |  |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Marks available |          |          |          |          |          |
|----------|--|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 8        |  |  |  | <p><b>Indicative content</b></p> <ul style="list-style-type: none"> <li>carbon dioxide level in the atmosphere increases</li> <li>this traps more heat in the atmosphere</li> <li>the Earth's temperature is rising</li> <li>melting polar ice / glaciers leads to increased sea levels and flooding</li> <li>extreme weather such as hotter summers / colder winters / more hurricanes</li> <li>habitat changes are caused by changes in climate</li> </ul> <p><b>5-6 marks</b><br/>Global warming clearly explained including the idea of increase in carbon dioxide trapping more heat; two consequences described<br/><i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i></p> <p><b>3-4 marks</b><br/>Global warming explained with link between carbon dioxide and increase in temperature; reference to two consequences<br/><i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i></p> <p><b>1-2 marks</b><br/>Reference to increasing temperature; reference to one consequence<br/><i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i></p> <p><b>0 marks</b><br/><i>No attempt made or no response worthy of credit.</i></p> | 6               |          |          | 6        |          |          |
|          |  |  |  | <b>Question 8 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>6</b>        | <b>0</b> | <b>0</b> | <b>6</b> | <b>0</b> | <b>0</b> |

## Common questions

| Question |     |      |    | Marking details                                                                                                                                                                        | Marks available |          |          |          |          |          |
|----------|-----|------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |      |    |                                                                                                                                                                                        | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 9/1      | (a) | (i)  | I  | (thermal) decomposition                                                                                                                                                                | 1               |          |          | 1        |          | 1        |
|          |     |      | II | $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$<br>award (1) for reactant<br>award (1) for products<br>ignore any attempt at balancing                                            |                 | 2        |          | 2        |          | 1        |
|          |     | (ii) |    | water                                                                                                                                                                                  | 1               |          |          | 1        |          |          |
|          | (b) |      |    | award (1) for any of following<br>making cement / concrete / plaster<br>making iron / steel<br>road building<br>in statues<br>neutral answers – building / building houses / buildings | 1               |          |          | 1        |          |          |
|          |     |      |    | <b>Question 9/1 total</b>                                                                                                                                                              | <b>3</b>        | <b>2</b> | <b>0</b> | <b>5</b> | <b>0</b> | <b>2</b> |

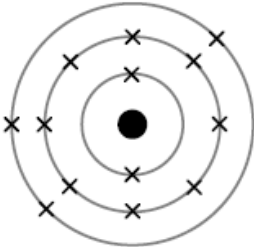
| Question            |     |  |  | Marking details                                                                                                                                                                                                                 | Marks available |     |     |       |       |      |
|---------------------|-----|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|                     |     |  |  |                                                                                                                                                                                                                                 | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 10/2                | (a) |  |  | <b>C</b> (1)<br><br>award (1) for any of following<br><br>$\frac{9}{15} = 0.6$<br>both <b>B</b> and <b>C</b> have $R_f$ of 0.6<br>both <b>B</b> and <b>C</b> have a dot at 9 cm<br><br>it is the highest dot (in <b>C</b> ) (1) |                 |     | 3   | 3     | 1     | 3    |
|                     | (b) |  |  | more soluble pigments move further up / more soluble pigments move faster (2)<br><br>pigments have different solubilities (1)<br><br>neutral answer – different $R_f$ values                                                    | 2               |     |     | 2     |       | 2    |
|                     | (c) |  |  | <b>B</b> (1)<br><br>award (1) for any of following<br>one of its dot has not moved / is still on the line<br>one of its dots has $R_f = 0$<br>pigment needs to be soluble to move up the paper                                  |                 |     | 2   | 2     |       | 2    |
|                     | (d) |  |  | 62 (2)<br><br>if incorrect award (1) for 36 or $\frac{12}{58}$                                                                                                                                                                  |                 | 2   |     | 2     | 2     |      |
| Question 10/2 total |     |  |  |                                                                                                                                                                                                                                 | 2               | 2   | 5   | 9     | 3     | 7    |

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                          | Marks available |          |          |          |          |          |
|----------|-----|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                          | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 11/3     | (a) |  |  | (surface of the) Earth cooled / temperature decreased (1)<br>(water vapour) condensed to form rivers/lakes/oceans (1)<br>award (1) each for any two of following<br>(carbon dioxide used in) photosynthesis / plants evolved<br>(carbon dioxide) locked in fossil fuels / rocks / shells<br>dissolved/absorbed in oceans | 4               |          |          | 4        |          |          |
|          | (b) |  |  | nitrogen $\Rightarrow$ 78% (1) accept 79 / 80<br>oxygen $\Rightarrow$ 21% (1) accept 20                                                                                                                                                                                                                                  | 2               |          |          | 2        |          |          |
|          |     |  |  | <b>Question 11/3 total</b>                                                                                                                                                                                                                                                                                               | <b>6</b>        | <b>0</b> | <b>0</b> | <b>6</b> | <b>0</b> | <b>0</b> |



## Higher Tier only questions

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                           | Marks available |          |          |          |          |          |
|----------|-----|------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                           | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 4        | (a) | (i)  |  | award (1) for sensible scale on y-axis e.g. 1 small square $\equiv$ 5 g<br><br>award (2) for 6 correct points (tolerance $\pm\frac{1}{2}$ square)<br>award (1) for any 4 or 5 correct points<br><br>award (1) for curve of best fit                                                                                                                                                       |                 | 3        | 1        | 4        | 4        |          |
|          |     | (ii) |  | 308 g (3)<br><br>accept any answer between 287 and 338 (based on $\pm\frac{1}{2}$ square tolerance for two readings from graph)<br><br>if incorrect award (2) for $169 - 46 = 123$ g<br><br>accept any answer between 115 and 135 (based on $\pm\frac{1}{2}$ square tolerance for two readings from graph)<br><br>award (1) for 46 g read from graph ( $\pm\frac{1}{2}$ square tolerance) |                 | 3        |          | 3        | 3        |          |
|          | (b) |      |  | ethanol and water have different boiling points / ethanol has a lower boiling point than water / water has a higher boiling point than ethanol (1)<br><br>award (1) for either of following<br>on heating, ethanol will evaporate first and go into the condenser<br>on heating, ethanol will evaporate at lower temperature and go into the condenser                                    | 2               |          |          | 2        |          | 2        |
|          |     |      |  | <b>Question 4 total</b>                                                                                                                                                                                                                                                                                                                                                                   | <b>2</b>        | <b>6</b> | <b>1</b> | <b>9</b> | <b>7</b> | <b>2</b> |

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks available |     |     |       |       |      |
|----------|-----|------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                             | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 5        | (a) |      |  |  <p>accept electrons shown as dots<br/>accept diagram with nucleus missing</p>                                                                                                                                                                                                                                                                             |                 | 1   |     | 1     |       |      |
|          | (b) | (i)  |  | <p><b>B</b> and <b>D</b> (1) must be correct to access second mark</p> <p>award (1) for either of following<br/>they have same number of protons but different number of neutrons<br/>they have same atomic number but different mass number</p> <p>ignore reference to electrons</p>                                                                                                                                                       | 1               | 1   |     | 2     |       |      |
|          |     | (ii) |  | <p><b>A</b> and <b>F</b> (1) must be correct to access second mark</p> <p>award (1) for any of following<br/>they have different numbers of protons and electrons<br/><b>A</b> has more electrons than protons and <b>F</b> has more protons than electrons</p> <p>neutral answer - they are <b>A</b><sup>-</sup> and <b>F</b><sup>+</sup><br/>do not award the mark if there is any suggestion that the number of neutrons is relevant</p> | 1               | 1   |     | 2     |       |      |
|          |     |      |  | Question 5 total                                                                                                                                                                                                                                                                                                                                                                                                                            | 2               | 3   | 0   | 5     | 0     | 0    |

| Question |                   |                  |                                               | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks available |                   |               |                                               |               |      |           |                  |                    |                |   |             |                  |                   |                 |   |  |  |   |  |   |
|----------|-------------------|------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------------------|---------------|-----------------------------------------------|---------------|------|-----------|------------------|--------------------|----------------|---|-------------|------------------|-------------------|-----------------|---|--|--|---|--|---|
|          |                   |                  |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | AO1             | AO2               | AO3           | Total                                         | Maths         | Prac |           |                  |                    |                |   |             |                  |                   |                 |   |  |  |   |  |   |
| 6        | (a)               | (i)              |                                               | <table><tr><td>Compound</td><td>Flame test colour</td><td>Symbol of ion</td><td>Observation on adding silver nitrate solution</td><td>Symbol of ion</td></tr><tr><td>S</td><td>brick red</td><td>Ca<sup>2+</sup></td><td>yellow precipitate</td><td>I<sup>-</sup></td></tr><tr><td>T</td><td>apple green</td><td>Ba<sup>2+</sup></td><td>white precipitate</td><td>Cl<sup>-</sup></td></tr></table> <p>award (1) mark for each correct answer</p> <p>do not accept ions with missing (or incorrect) charges but penalise once only e.g.</p> <p>award (2) if all three ions identified but no charges given</p> <p>award (1) if two ions identified but no charges given</p> | Compound        | Flame test colour | Symbol of ion | Observation on adding silver nitrate solution | Symbol of ion | S    | brick red | Ca <sup>2+</sup> | yellow precipitate | I <sup>-</sup> | T | apple green | Ba <sup>2+</sup> | white precipitate | Cl <sup>-</sup> | 4 |  |  | 4 |  | 4 |
| Compound | Flame test colour | Symbol of ion    | Observation on adding silver nitrate solution | Symbol of ion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                 |                   |               |                                               |               |      |           |                  |                    |                |   |             |                  |                   |                 |   |  |  |   |  |   |
| S        | brick red         | Ca <sup>2+</sup> | yellow precipitate                            | I <sup>-</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                 |                   |               |                                               |               |      |           |                  |                    |                |   |             |                  |                   |                 |   |  |  |   |  |   |
| T        | apple green       | Ba <sup>2+</sup> | white precipitate                             | Cl <sup>-</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                 |                   |               |                                               |               |      |           |                  |                    |                |   |             |                  |                   |                 |   |  |  |   |  |   |
|          |                   | (ii)             |                                               | <p>award (1) for any of following</p> <p>the yellow would hide the white precipitate</p> <p>it would look (pale) yellow (like iodide) / cream (like bromide)</p> <p>there would be a mixture of two precipitates</p> <p>it would not be possible to distinguish between the colours</p>                                                                                                                                                                                                                                                                                                                                                                                     |                 |                   | 1             | 1                                             |               | 1    |           |                  |                    |                |   |             |                  |                   |                 |   |  |  |   |  |   |
|          | (b)               |                  |                                               | <p>Ag<sup>+</sup>(aq) + Cl<sup>-</sup>(aq) → AgCl(s)</p> <p>award (1) for ions</p> <p>award (1) for product formula</p> <p>award (1) for state symbols – can only be awarded if ions and product are correct</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                 | 3                 |               | 3                                             |               |      |           |                  |                    |                |   |             |                  |                   |                 |   |  |  |   |  |   |
|          |                   |                  |                                               | Question 6 total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 4               | 3                 | 1             | 8                                             | 0             | 5    |           |                  |                    |                |   |             |                  |                   |                 |   |  |  |   |  |   |

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                         | Marks available |          |          |          |          |          |
|----------|-----|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                         | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 7        | (a) |  |  | <b>4</b>                                                                                                                                                                                                                                                                                                                |                 |          | 1        | 1        | 1        |          |
|          | (b) |  |  | <b>Y (1)</b><br>award (1) for either of following<br>volume of gas produced / rate of reaction increases with temperature then decreases<br>it works best / has an optimum temperature at around 40°C<br>enzymes are denatured at 40°C / at higher temperatures (1)<br>neutral answers – broken down / damaged / killed | 1               |          | 2        | 3        |          |          |
|          | (c) |  |  | <b>C</b>                                                                                                                                                                                                                                                                                                                |                 |          | 1        | 1        |          |          |
|          |     |  |  | <b>Question 7 total</b>                                                                                                                                                                                                                                                                                                 | <b>1</b>        | <b>0</b> | <b>4</b> | <b>5</b> | <b>1</b> | <b>0</b> |

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                     | Marks available |          |          |          |          |          |
|----------|-----|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                     | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 8        | (a) |  |  | <p><b>A</b> is permanent hard water (1)</p> <p>it is not softened by boiling (only by ion exchange) / boiling has no effect on the volume of soap needed (1)</p> <p><b>B</b> contains both temporary and permanent hard water (1)</p> <p>as it is partly softened by boiling <u>and</u> further softened by ion exchange / less soap needed after boiling and less again after ion exchange (1)</p> | 2               |          | 2        | 4        |          | 4        |
|          | (b) |  |  | <p><math>\text{Na}_2\text{CO}_3 + \text{MgCl}_2 \rightarrow 2\text{NaCl} + \text{MgCO}_3</math></p> <p>award (1) for reactants<br/>award (1) for products<br/>award (1) for balancing<br/>- can only be awarded if reactants and products are correct</p> <p>accept multiples of correct balancing<br/>ignore state symbols</p>                                                                     |                 | 3        |          | 3        |          |          |
|          |     |  |  | <b>Question 8 total</b>                                                                                                                                                                                                                                                                                                                                                                             | <b>2</b>        | <b>3</b> | <b>2</b> | <b>7</b> | <b>0</b> | <b>4</b> |

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marks available |     |     |       |       |      |
|----------|-----|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 9        | (a) |  |  | <p><b>Indicative content</b></p> <ul style="list-style-type: none"> <li>lithium fizzes and moves around the surface of the water</li> <li>sodium moves faster on the surface, fizzes more and melts into a ball</li> <li>potassium reacts more vigorously again, melts into a ball and ignites producing a lilac flame</li> <li>reactions more vigorous on moving down the group</li> <li>outer electron is lost during the reaction</li> <li>lost more easily on moving down the group because it is further away from the nucleus / attraction between the nucleus and the outer electron decreases</li> </ul> <p><b>5-6 marks</b><br/>Detailed description of reactions; explanation of relative ease of loss of outer electron<br/><i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i></p> <p><b>3-4 marks</b><br/>Basic description of reactions; reference to loss of outer electron<br/><i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i></p> <p><b>1-2 marks</b><br/>Basic description of some reactions<br/><i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i></p> <p><b>0 marks</b><br/><i>No attempt made or no response worthy of credit.</i></p> | 6               |     |     | 6     |       | 3    |

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks available |          |          |           |          |          |
|----------|-----|------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
|          | (b) | (i)  |  | 18.8 g (3)<br><br>if answer incorrect credit each correct step in one of two possible methods (ecf possible throughout)<br><br>method 1<br><br>$n(K) = \frac{15.6}{39} = 0.4 \text{ mol} \quad (1)$<br><br>$n(K_2O) = \frac{0.4}{2} = 0.2 \text{ mol} \quad (1)$<br><br>$\text{mass } K_2O = 0.2 \times 94 = 18.8 \text{ g} \quad (1)$<br><br>method 2<br><br>$M_r(K_2O) = 94 / \text{mass of } 156 \text{ (for K)} (1)$<br><br>(156 g K produces) 188 g $K_2O$ (1)<br><br>15.6 g K produces 18.8 g $K_2O$ (1) |                 | 3        |          | 3         | 3        |          |
|          |     | (ii) |  | $3.0 \times 10^{22}$ (2) accept $3 \times 10^{22}$<br><br>if answer incorrect award 1 mark for $0.30 \times 10^{23}$                                                                                                                                                                                                                                                                                                                                                                                           |                 | 2        |          | 2         | 2        |          |
|          |     |      |  | <b>Question 9 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>6</b>        | <b>5</b> | <b>0</b> | <b>11</b> | <b>5</b> | <b>3</b> |

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                            | Marks available |          |          |          |          |          |
|----------|-----|------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                            | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 10       | (a) | (i)  |  | <p><b>C (1)</b></p> <p>reactivity of halogens decreases down the table / chlorine is the most reactive/ iodine is the least reactive (1)</p> <p>award (1) for any of following<br/> chlorine displaces bromine and iodine<br/> iodine does not displace bromine or chlorine<br/> chlorine reacts with sodium bromide and sodium iodide<br/> iodine does not react with sodium bromide or sodium iodide</p> | 2               |          | 1        | 3        |          | 3        |
|          |     | (ii) |  | <p><math>\text{Cl}_2 + 2\text{NaI} \rightarrow 2\text{NaCl} + \text{I}_2</math></p> <p>award (1) for reactants<br/> award (1) for products<br/> award (1) for balancing<br/> - can only be awarded if reactants and products are correct<br/> ignore state symbols</p> <p>accept ionic equation</p> <p><math>\text{Cl}_2 + 2\text{I}^- \rightarrow 2\text{Cl}^- + \text{I}_2</math></p>                    |                 |          |          | 3        |          |          |
|          | (b) |      |  | <p><math>n(\text{Fe}) = \frac{7}{56} = 0.125</math> (1)</p> <p><math>n(\text{Br}) = \frac{30}{80} = 0.375</math> (1)</p> <p>ratio 1:3 therefore <math>\text{FeBr}_3</math> (1)</p> <p><b>working must be shown</b></p>                                                                                                                                                                                     |                 | 3        |          | 3        | 2        |          |
|          |     |      |  | <b>Question 10 total</b>                                                                                                                                                                                                                                                                                                                                                                                   | <b>2</b>        | <b>6</b> | <b>1</b> | <b>9</b> | <b>2</b> | <b>3</b> |



| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                      | Marks available |          |          |          |          |          |
|----------|-----|------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                      | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 11       | (a) |      |  | <p>1.52 cm<sup>3</sup>/s (2)                      accept 1.5</p> <p>if incorrect award (1) for either of following<br/>(58 – 20) and (30 – 5)<br/>38 and 25</p> <p>ecf possible if one value read incorrectly from graph</p>                                                                                                                         |                 | 2        |          | 2        | 2        |          |
|          | (b) | (i)  |  | <p>line steeper than original line (1)</p> <p>line finishing at 90 cm<sup>3</sup> (1)</p>                                                                                                                                                                                                                                                            |                 |          | 2        | 2        |          | 2        |
|          |     | (ii) |  | <p>award (1) each for any two of following</p> <ul style="list-style-type: none"> <li>• greater surface area at the start</li> <li>• more collisions per unit time / more frequent collisions</li> <li>• produces 50% more gas as mass is 50% more</li> <li>• carbonate is the limiting factor / reaction stops when carbonate is used up</li> </ul> | 2               |          |          | 2        |          |          |
|          |     |      |  | <b>Question 11 total</b>                                                                                                                                                                                                                                                                                                                             | <b>2</b>        | <b>2</b> | <b>2</b> | <b>6</b> | <b>2</b> | <b>2</b> |

## FOUNDATION TIER

## SUMMARY OF MARKS ALLOCATED TO ASSESSMENT OBJECTIVES

| Question | AO1 | AO2 | AO3 | TOTAL MARK | MATHS | PRAC |
|----------|-----|-----|-----|------------|-------|------|
| 1        | 4   | 3   | 0   | 7          | 1     | 4    |
| 2        | 0   | 3   | 0   | 3          | 0     | 0    |
| 3        | 1   | 5   | 0   | 6          | 0     | 1    |
| 4        | 5   | 0   | 3   | 8          | 1     | 0    |
| 5        | 1   | 8   | 3   | 12         | 5     | 3    |
| 6        | 2   | 6   | 0   | 8          | 0     | 0    |
| 7        | 2   | 3   | 5   | 10         | 3     | 5    |
| 8        | 6   | 0   | 0   | 6          | 0     | 0    |
| 9        | 3   | 2   | 0   | 5          | 0     | 2    |
| 10       | 2   | 2   | 5   | 9          | 3     | 7    |
| 11       | 6   | 0   | 0   | 6          | 0     | 0    |
| TOTAL    | 32  | 32  | 16  | 80         | 13    | 22   |

## HIGHER TIER

## SUMMARY OF MARKS ALLOCATED TO ASSESSMENT OBJECTIVES

| Question | AO1 | AO2 | AO3 | TOTAL MARK | MATHS | PRAC |
|----------|-----|-----|-----|------------|-------|------|
| 1        | 3   | 2   | 0   | 5          | 0     | 2    |
| 2        | 2   | 2   | 5   | 9          | 3     | 7    |
| 3        | 6   | 0   | 0   | 6          | 0     | 0    |
| 4        | 2   | 6   | 1   | 9          | 7     | 2    |
| 5        | 2   | 3   | 0   | 5          | 0     | 0    |
| 6        | 4   | 3   | 1   | 8          | 0     | 5    |
| 7        | 1   | 0   | 4   | 5          | 1     | 0    |
| 8        | 2   | 3   | 2   | 7          | 0     | 4    |
| 9        | 6   | 5   | 0   | 11         | 5     | 3    |
| 10       | 2   | 6   | 1   | 9          | 2     | 3    |
| 11       | 2   | 2   | 2   | 6          | 2     | 2    |
| TOTAL    | 32  | 32  | 16  | 80         | 20    | 28   |